

The Cocoon Portal

More than the Portlet API (JSR 168)

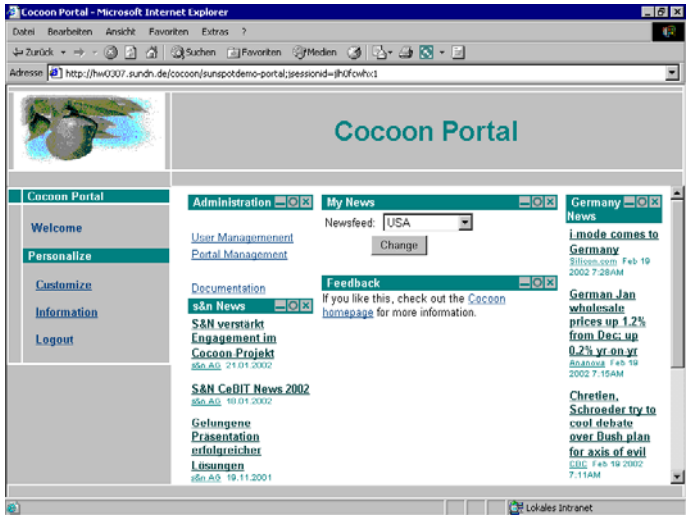
Carsten Ziegeler
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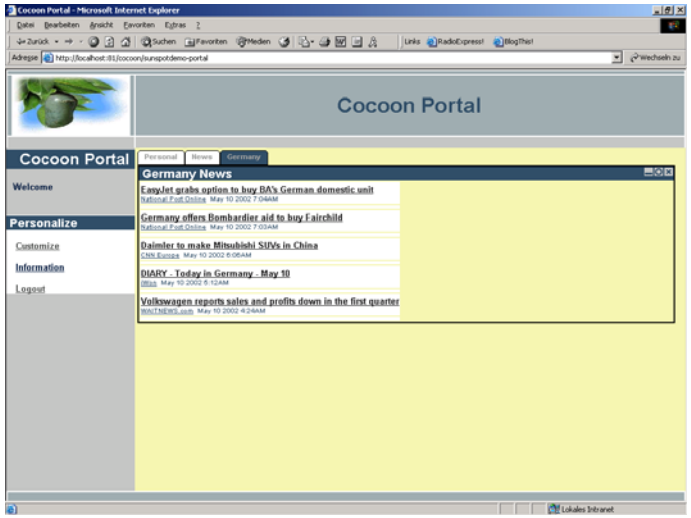
Competence Center Open Source
S&N AG, Germany



Agenda

- **Portal Basics**
- **Cocoon Introduction**
- **Opening the Portal**
- **The Future**

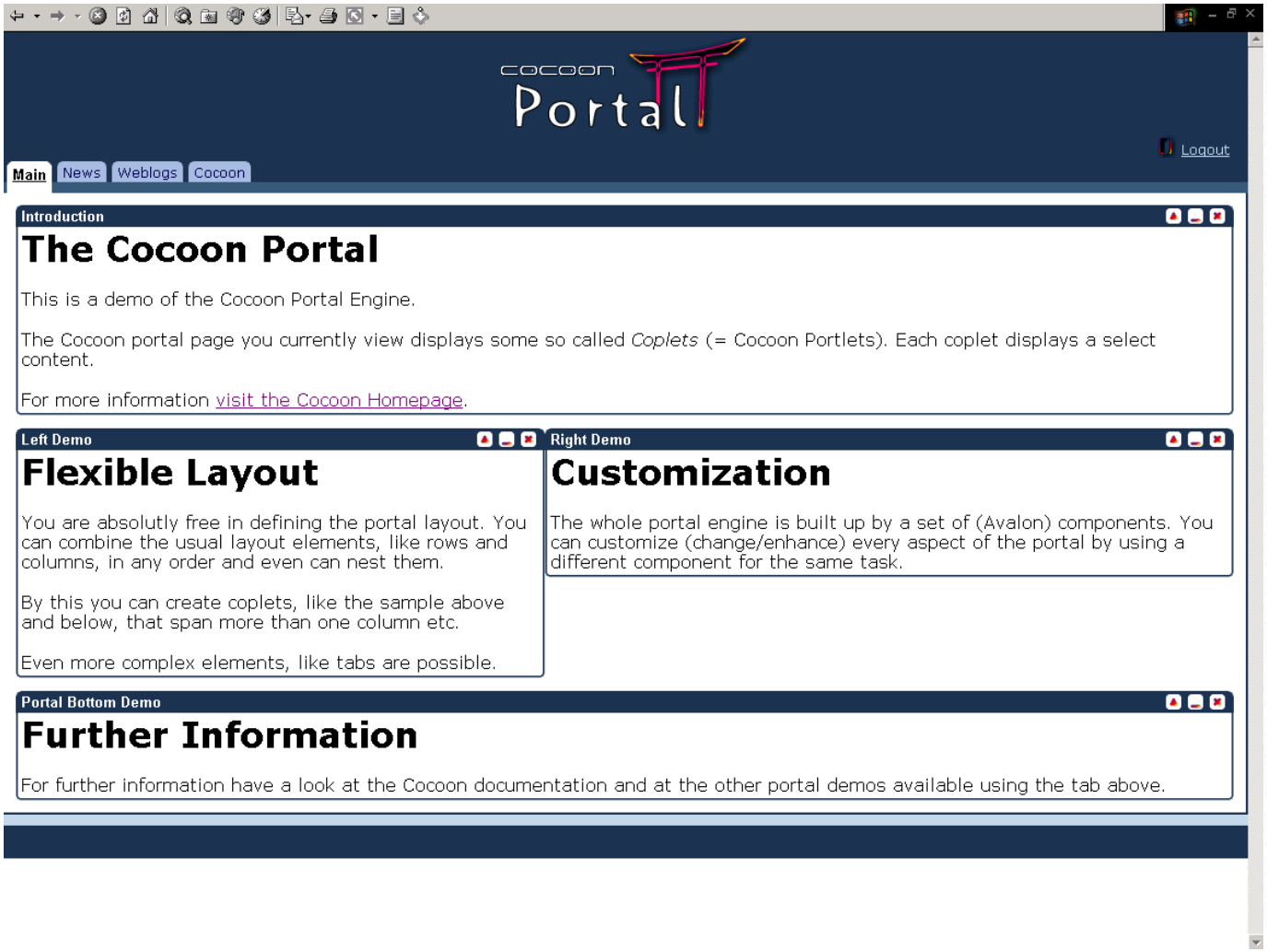




What is a Portal?

- Web Based Application
  - Personalization
  - Individualization
  - Content Aggregation
    - Using Portlets
  - Single Sign On

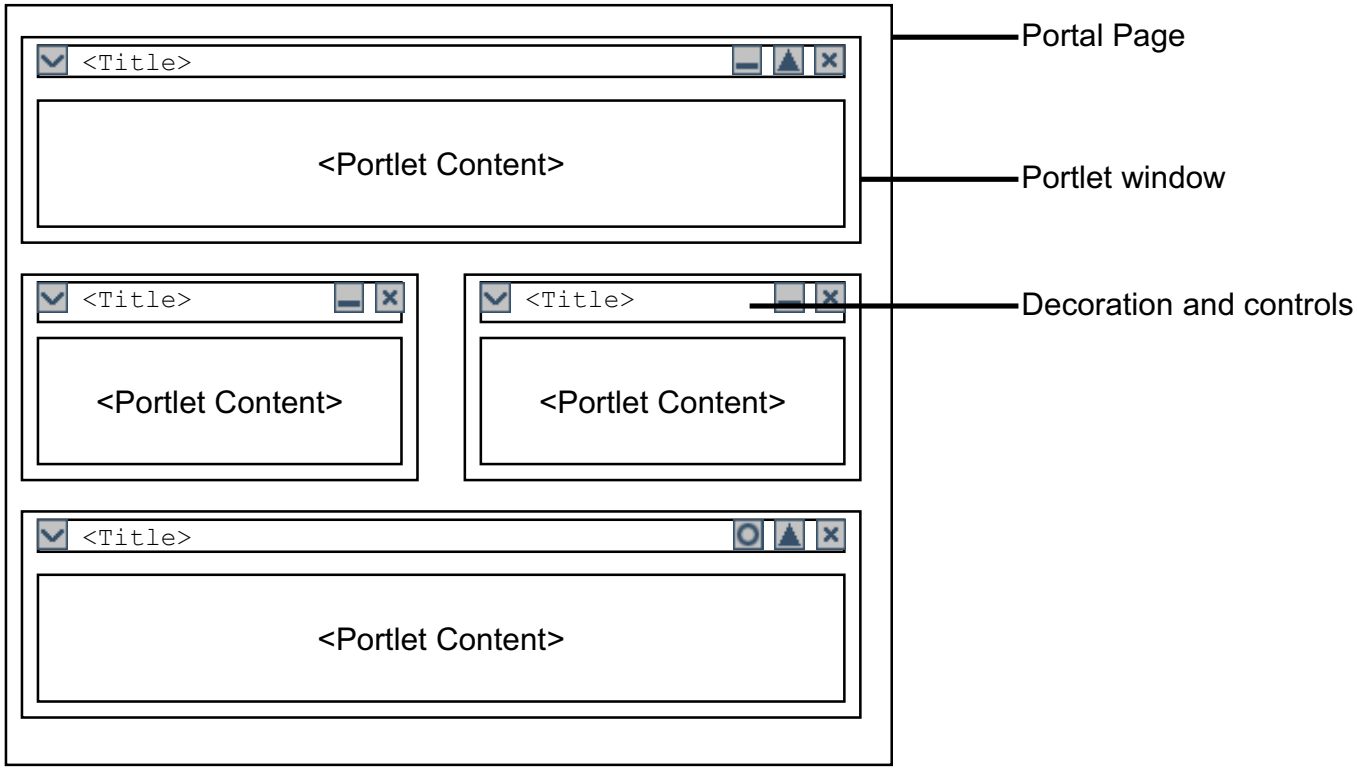
What is a Portal?



What is a Portlet?

- Web Component
  - Generates (dynamic) Content
    - News
    - Links
    - Complete Web Application
    - …
  - Managed by a Portlet Container

A Portal Page Sample



What is a Portlet Container?

- Runs Portlets
- Manages Portlets
- Persistent Storage for Preferences

What is a Portal Server?

- Web Server
- With integrated Portlet Container
- Runs the developed Portal (Application)

The JSR 168 – Portlet API

- Java API (based on J2EE 1.3)
  - Portlet Development
  - User Preferences
  - Localization
- Similar to Servlet API
  - Request-Response Cycle
  - Own Deployment Descriptor
- Portlet Container extends Servlet Container
  - Servlet Specification 2.3
  - **Not** covered in the JSR
- Reference Implementation: Apache Pluto

Entering Cocoon…

- The Cocoon Portal is more than just the Portlet API
  - An integrated portal server
  - A coplet container
    - coplet = **Co**coon **P**ort**let**
    - Integrated portlet container
  - Highly extensible

Cocoon Basics

XML Publishing



Apache Cocoon

- Apache Open Source project
  - http://cocoon.apache.org
- Started in 1999 by Stefano Mazzocchi
  - Original goal: XML Publishing Framework
  - Now a thriving community
- XML Publishing Platform + Web Application Framework
  - Makes extensive use of XML and XSLT
- Aim: To Separate Concerns (SoC)
  - Site Administrator, Content Deployer, Layout Deployer
- Written in Java

Cocoon Overview

- Dynamic Document Generation
  - Based on XML/XSLT
  - But not limited to
- Runs in different environments
  - Servlet Engine
  - Command Line
- Used for various application scenarios
  - Web Sites
  - Web Publishing
  - XML Portals
  - XML Processing Systems
  - …

Cocoon – Major Advantages

- Extensible Architecture
  - Component Orientated
  - Based on Apache Avalon Framework
  - Already integrates other projects
    - Xalan, Xerces
    - FOP, Batik, POI
  - Add new/own components (if required)
    - Seamless Integration

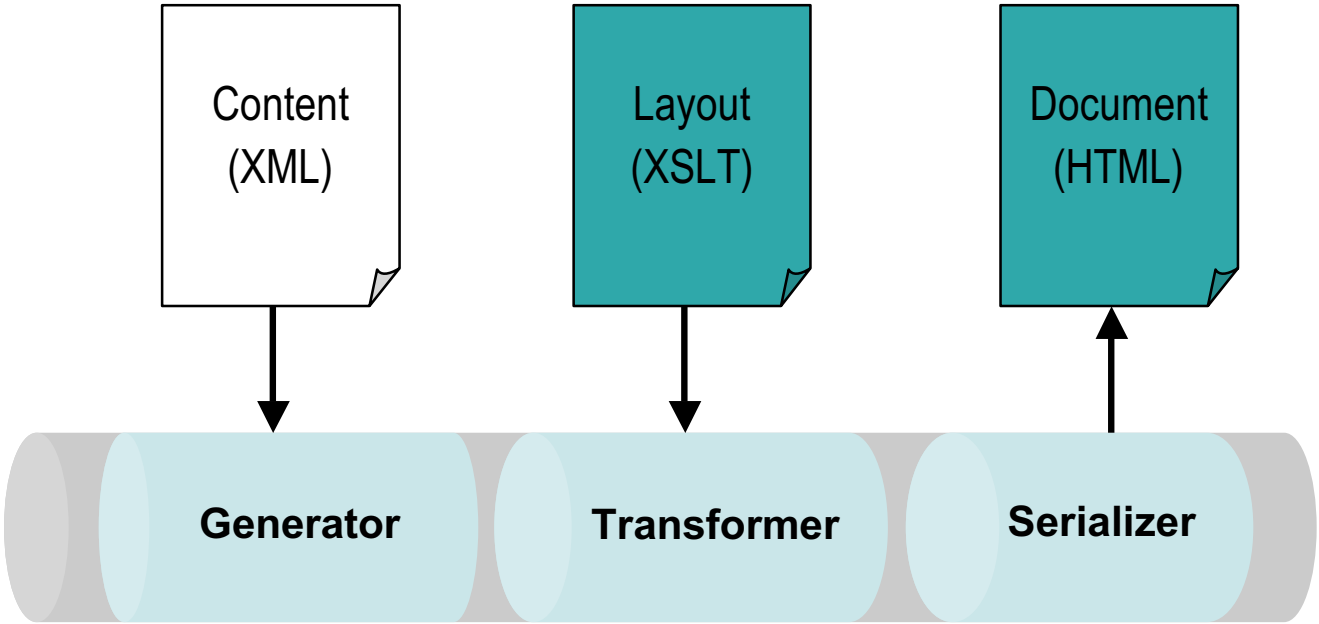
Cocoon – Major Advantages

- Document generation on the server
  - Powerful Process Description
  - Logical names used „http://myserver/cocoon/helloworld“
- Flexible Data Integration
  - XML files, XML over HTTP
  - Databases, LDAP, SAP
  - …
- Flexible Publishing to different formats using XSLT
  - HTML, WML, XML
  - PDF, SVG, PS, Office Documents
  - …

Cocoon – Documentation Generation

- Dynamic Document Generation
  - Based on Request-Response-Cycle
  - By defining a processing pipeline per document
    - Getting Content
    - Adding Layout
    - Sending Response
  - Extensible by Logic and Flow

Building a Pipeline



Simple Concept – Hugh Impact





The Cocoon Portal

Introduction



History of the Cocoon Portal

- Started in September 2000
  - Commercial product developed by S&N
  - Based on Cocoon
  - Additional components and project specific additions
  - CMS, Tools, documentation, trainings
- Completed in mid 2001
  - sunRise and sunSpot
  - Used in various projects
- Donated to the Cocoon project in February 2002
  - Used around the world
- Started new Portal Engine in the end of 2002
  - The usual advantages

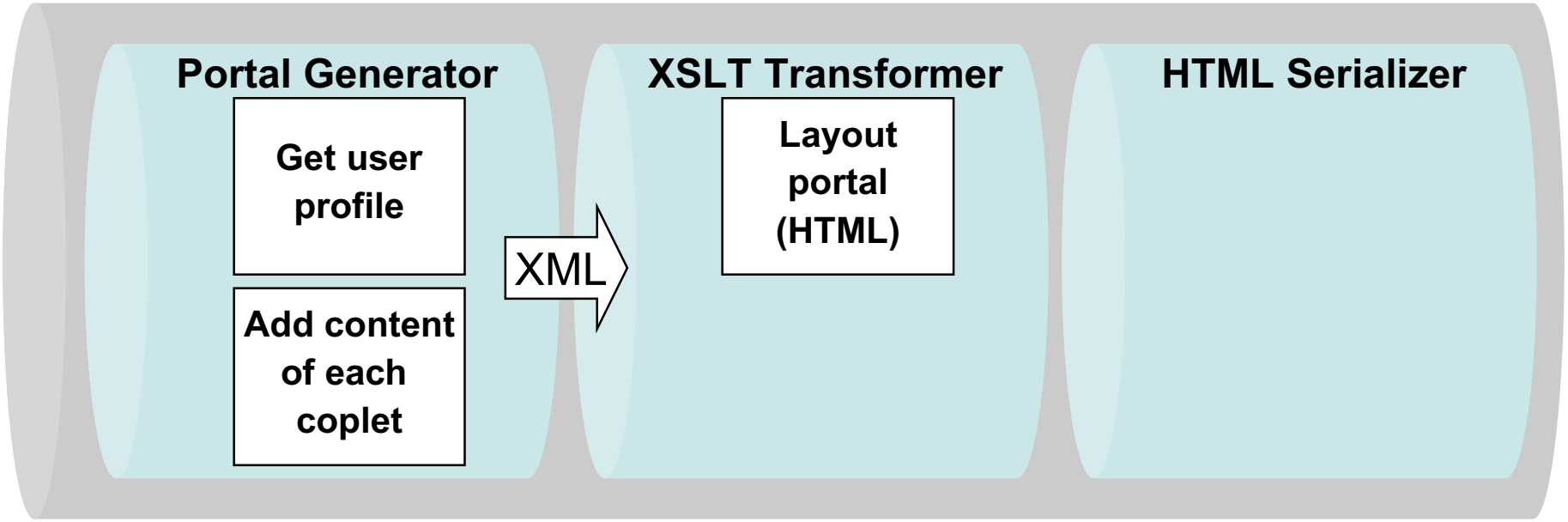
Common Requirements…

- Integration of different data sources
  - Static sources (HTML, XML, Office Documents…)
  - Dynamic sources (CMS, Archives…)
  - Databases (SQL DB, XML DB, LDAP…)
  - Complex Applications
- Single Sign-On
- Personalization
- Individualization
- Multi Channel
  - PCs (HTML, XML)
  - Mobile, Organizer (WML)
  - Documents (PDF, Office Documents)
  - Email
  - Applications

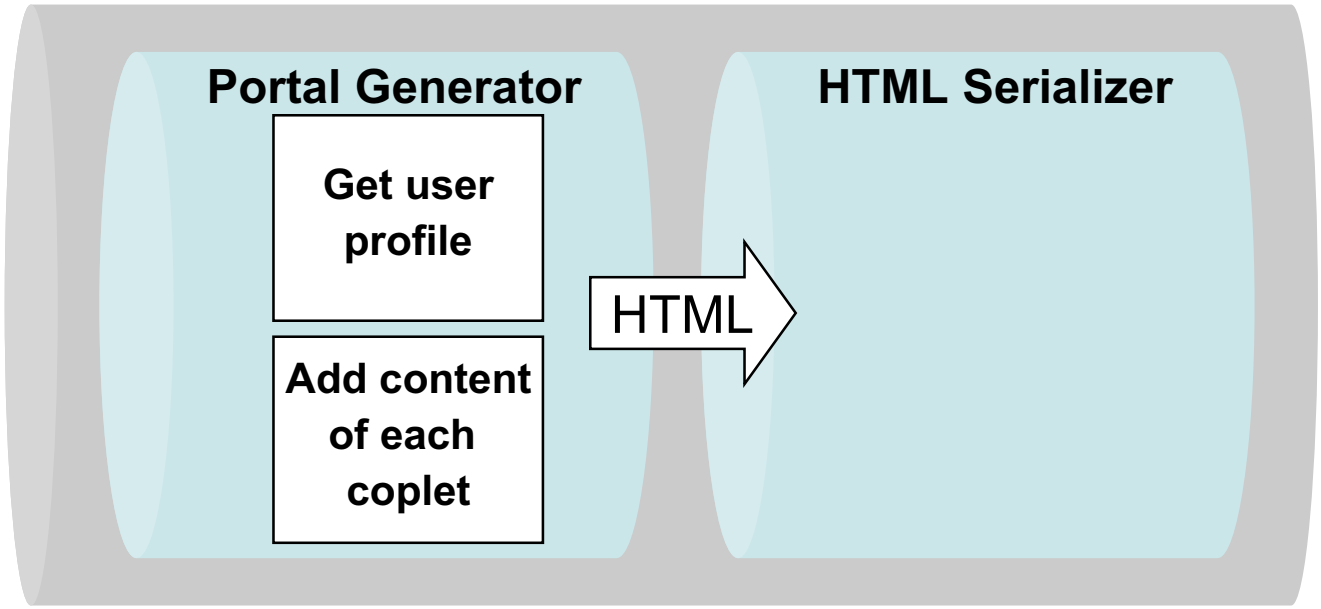
…fulfilled by Cocoon

- Advantages of the pipeline concept
  - Different Generators for retrieving data
  - Different Transformers for manipulating/augmenting data
  - Stylesheet Transformation
    - for various output formats
    - for customized pages
- Logic
  - Format Detection
  - Page Flow
  - Actions

Portal Pipeline



Portal Pipeline (alternative)



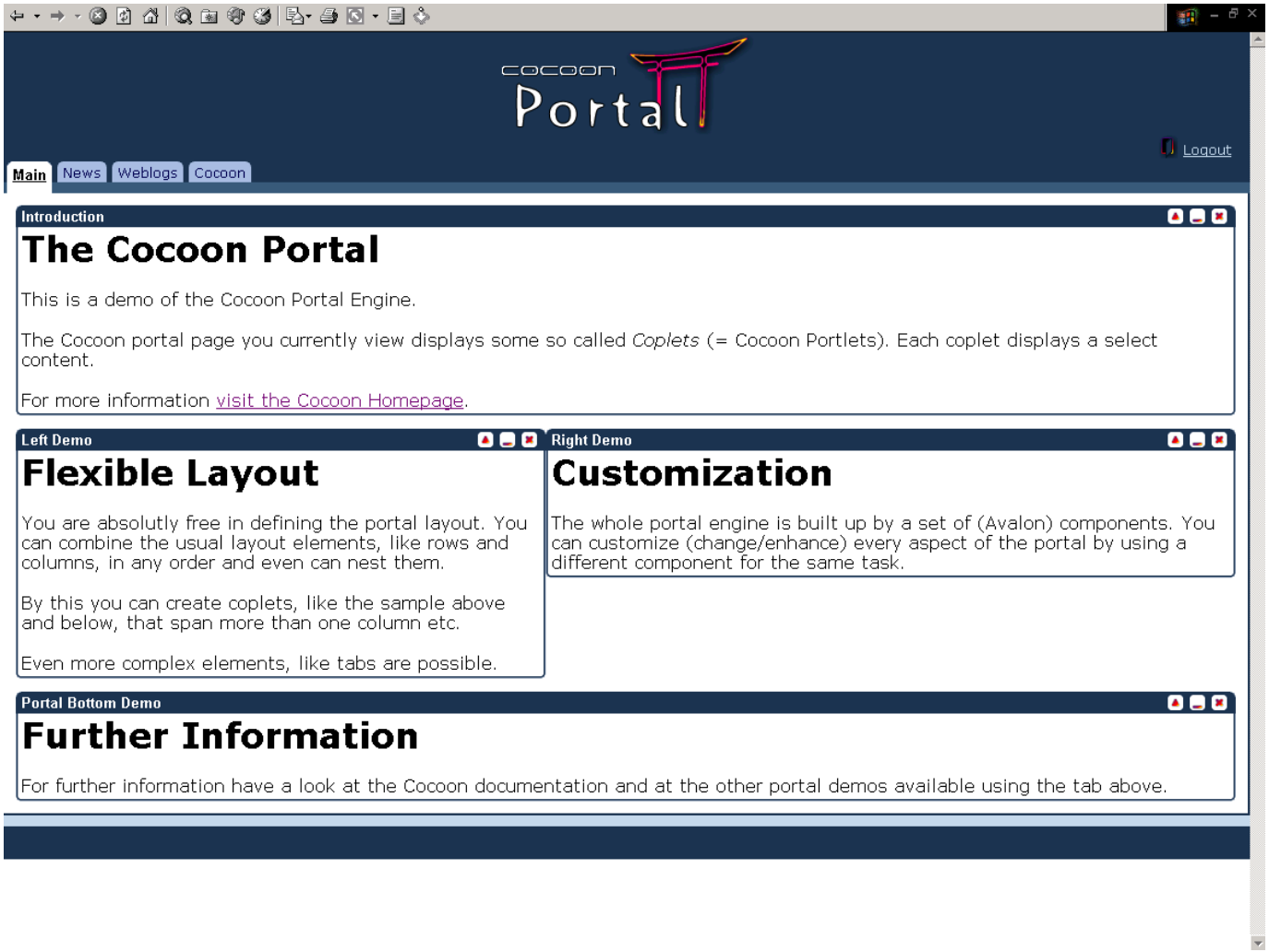
Generating the Portal View

- Getting the Profile
  - Layout
  - User preferences
- Rendering the View
  - Drawing the layout
- Aggregating Content
  - Include content of selected coplets

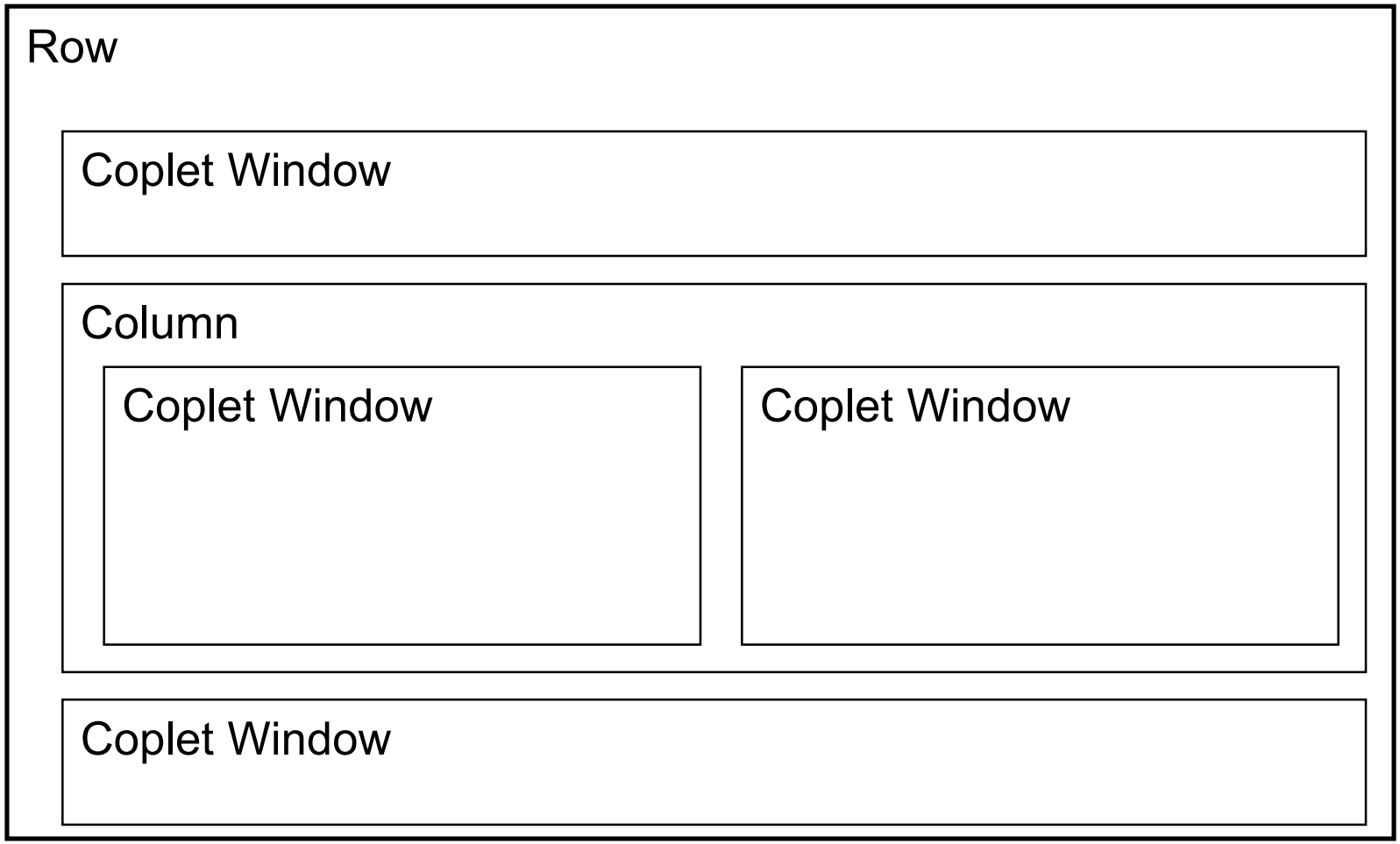
Defining The Portal Layout

- Hierarchical Tree of
  - Composite Layout Objects
    - Row, Column
    - Tab
    - …
  - Single Layout Objects
    - Coplet Window
    - Frame
    - Link
    - …
  - Deep nesting possible!

Defining The Portal Layout



Defining The Portal Layout



Defining The Portal Layout

```
<composite-layout name="row">
  <item>
    <coplet-layout name="coplet">
        <coplet-instance-data>First</coplet-instance-data>
    </coplet-layout>
  </item>
  <item>
    <composite-layout name="column">
      <item>
        <coplet-layout name="coplet">
          <coplet-instance-data>Left</coplet-instance-data>
        </coplet-layout>
      </item>
      <item>
        <coplet-layout name="coplet">
          <coplet-instance-data>Right</coplet-instance-data>
        </coplet-layout>
      </item>
    </composite-layout>
  </item>
  <item>
    <coplet-layout name="coplet">
      <coplet-instance-data>Bottom</coplet-instance-data>
    </coplet-layout>
  </item>
</composite-layout>
```

Rendering The Portal Layout

- Configured renderer for each layout
  - XML, HTML, WML, …
- Renderers traverse the tree
- Placeholder for each coplet

```
<table>
  <tr><td>
    <coplet-layout name="coplet">

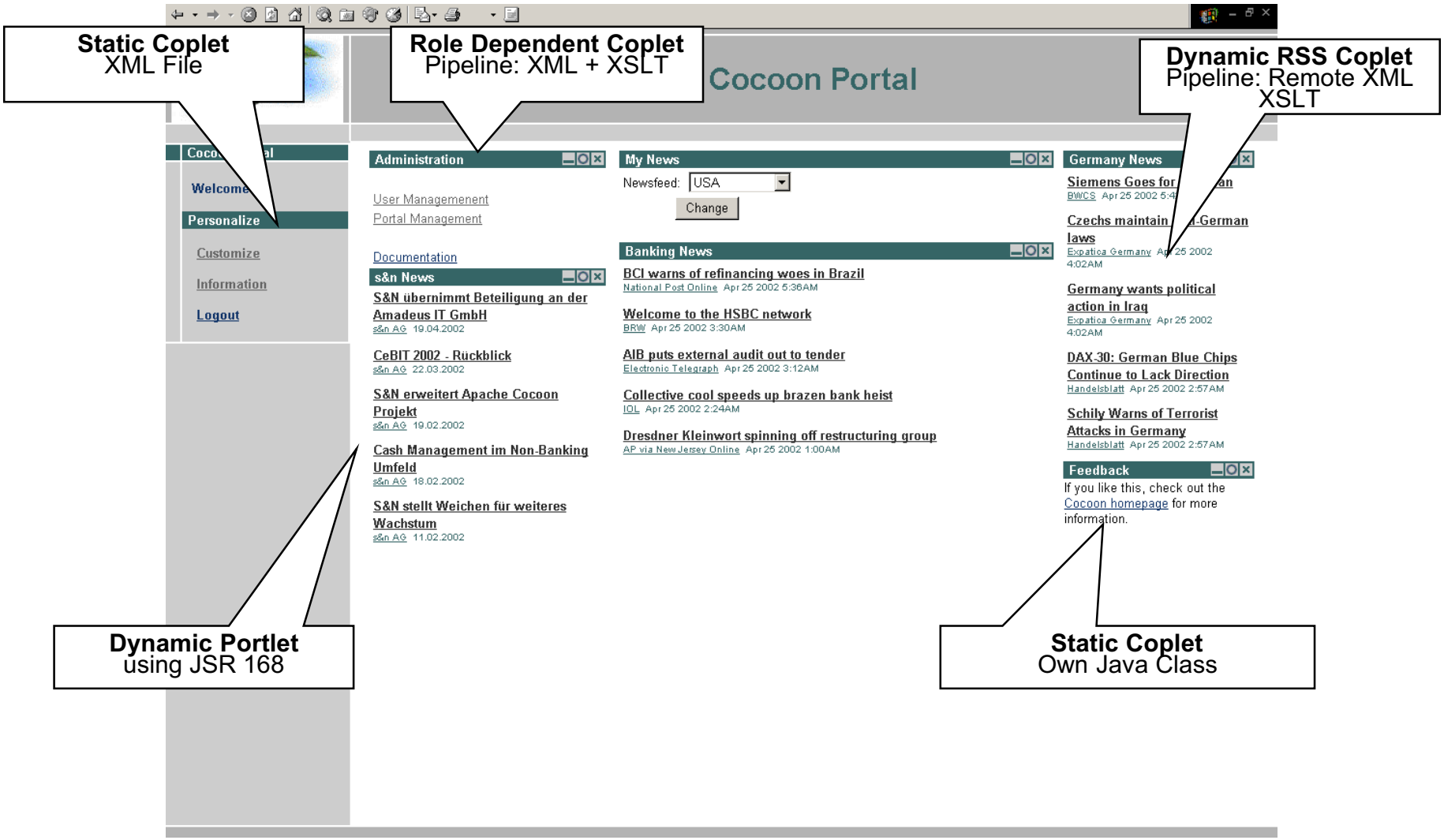
      <coplet-instance-data>First</coplet-instance-data>

    </coplet-layout>
  </td></tr>
  ...
</table>
```

Getting Content from Coplets

- Different types of coplets
  - Static
  - URI based
  - Pipeline based
  - JSR 168 portlets
  - Custom

The Cocoon Portal



Getting Content from Coplets

- Definition of coplets (_ classes)
  - Unique id
  - Required type
  - Configuration

```
<coplet-data id="CZ Weblog" name="standard">
   <title>CZ's Weblog</title>
   <coplet-base-data>URICoplet</coplet-base-data>
   <uri>cocoon:/pipelines/portal/weblog</uri>
</coplet-data>
```

ID

Type

Optional Configuration

Getting Content from Coplets

- Usage of coplets (_ instances)
  - Unique id
  - Required coplet
  - Optional instance configuration

```
<coplet-instance-data id="First" name="standard">

    <coplet-data>CZ Weblog</coplet-data>

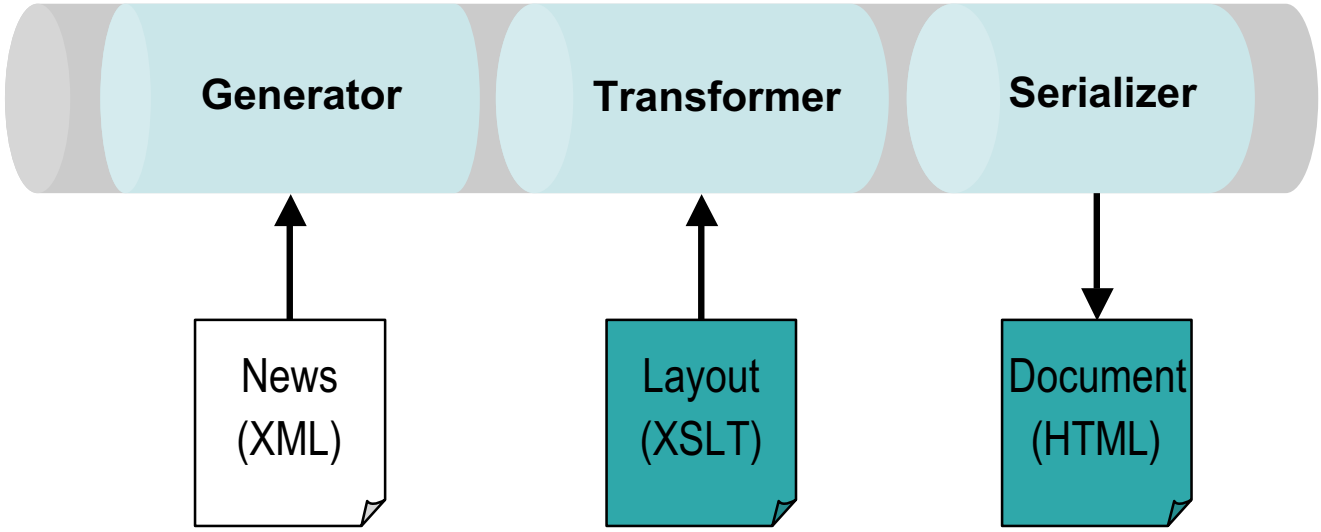
</coplet-instance-data>
```

ID

Coplet

Coplet Sample

- Reads news from RSS feed
  - Example: http://radio.weblogs.com/0107211/rss.xml
- Transforms news to HTML
- Uses a simple Cocoon pipeline



Cocoon Portal

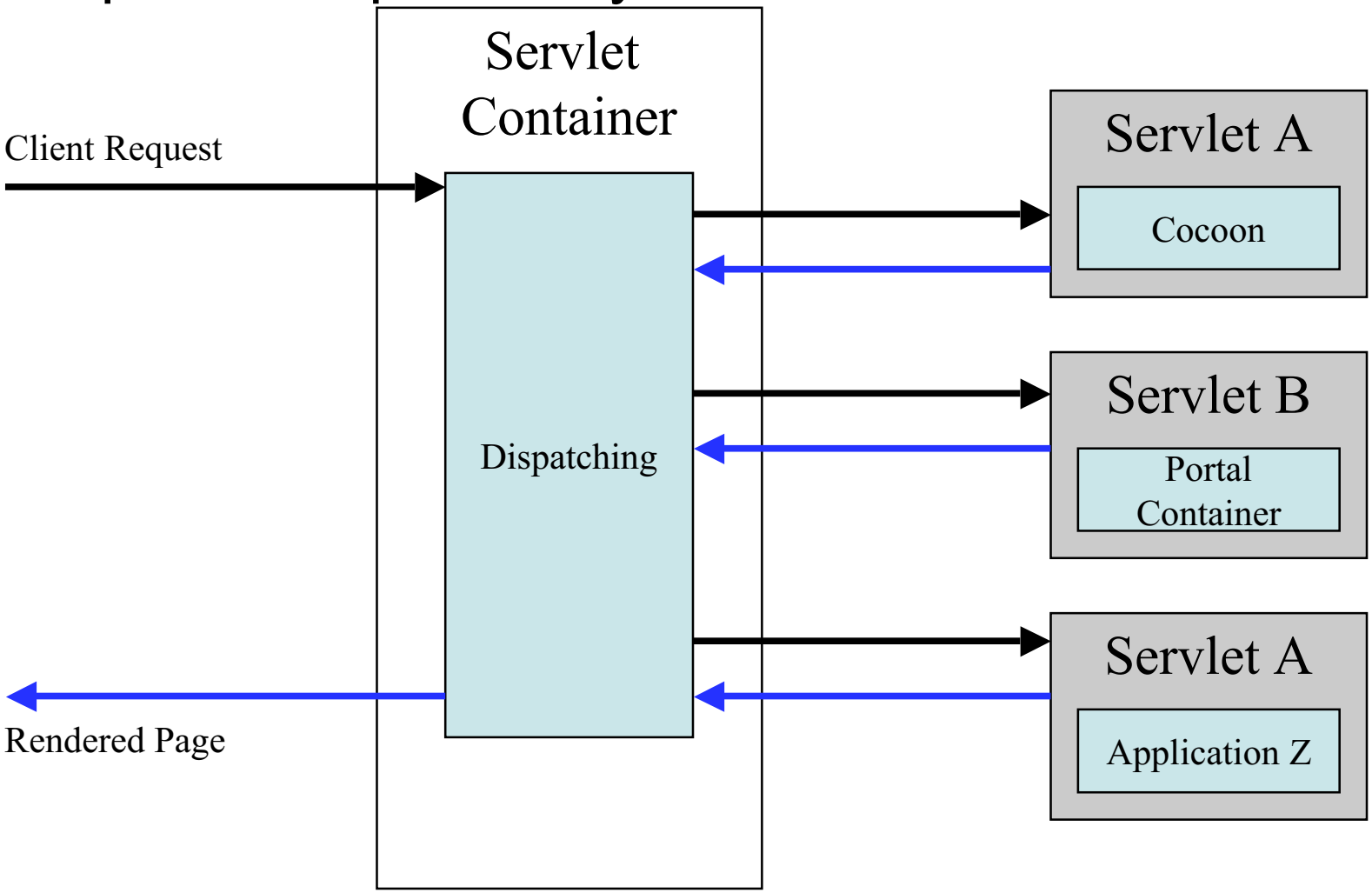
- Definition and configuration in XML
  - Layout (ordering) separated from content
  - Using existing types
  - Requires no Java coding
- Predefined coplet types
  - Simple usage of Cocoon for delivering content
  - Requires no Java coding

The Cocoon Portal

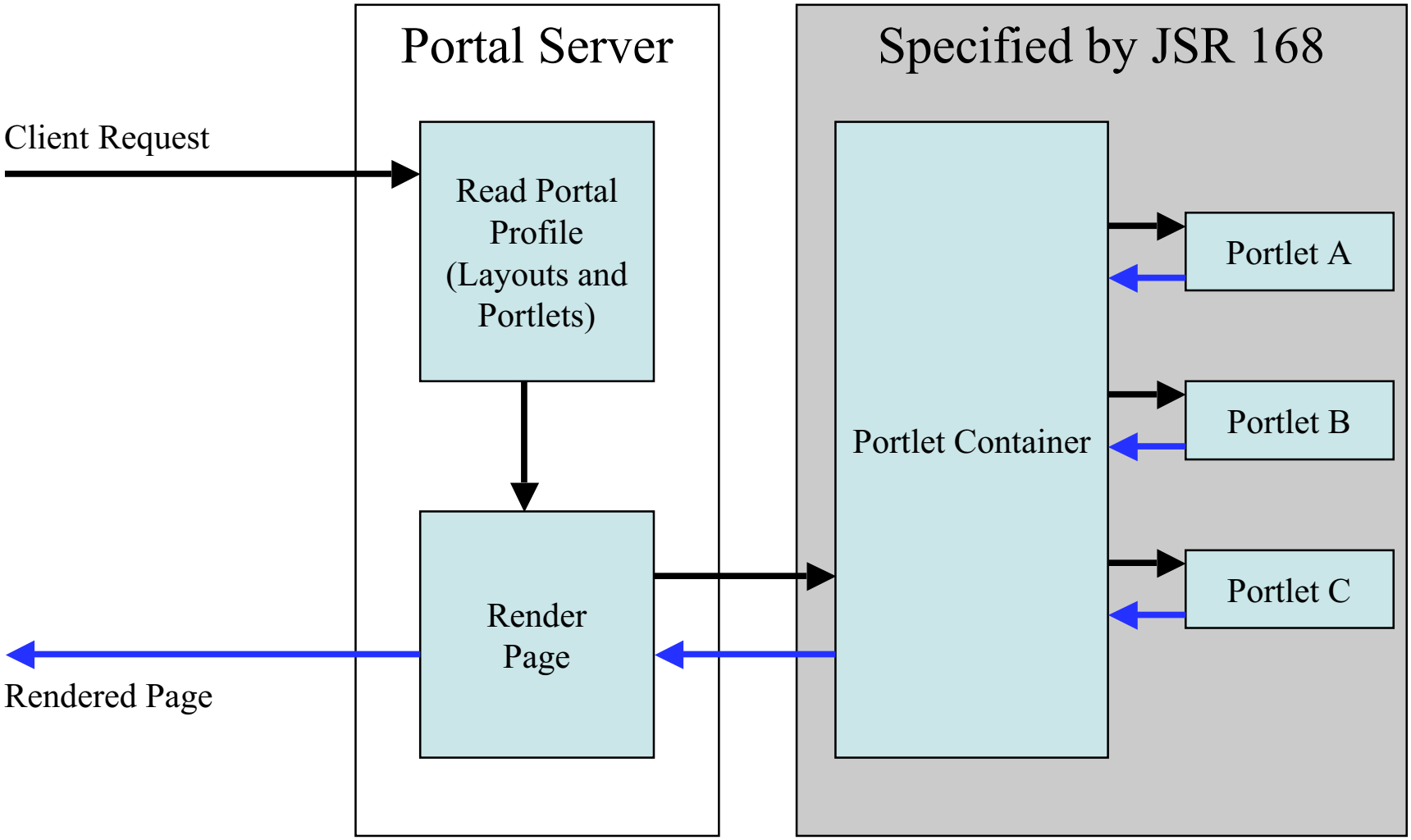
The Big Picture



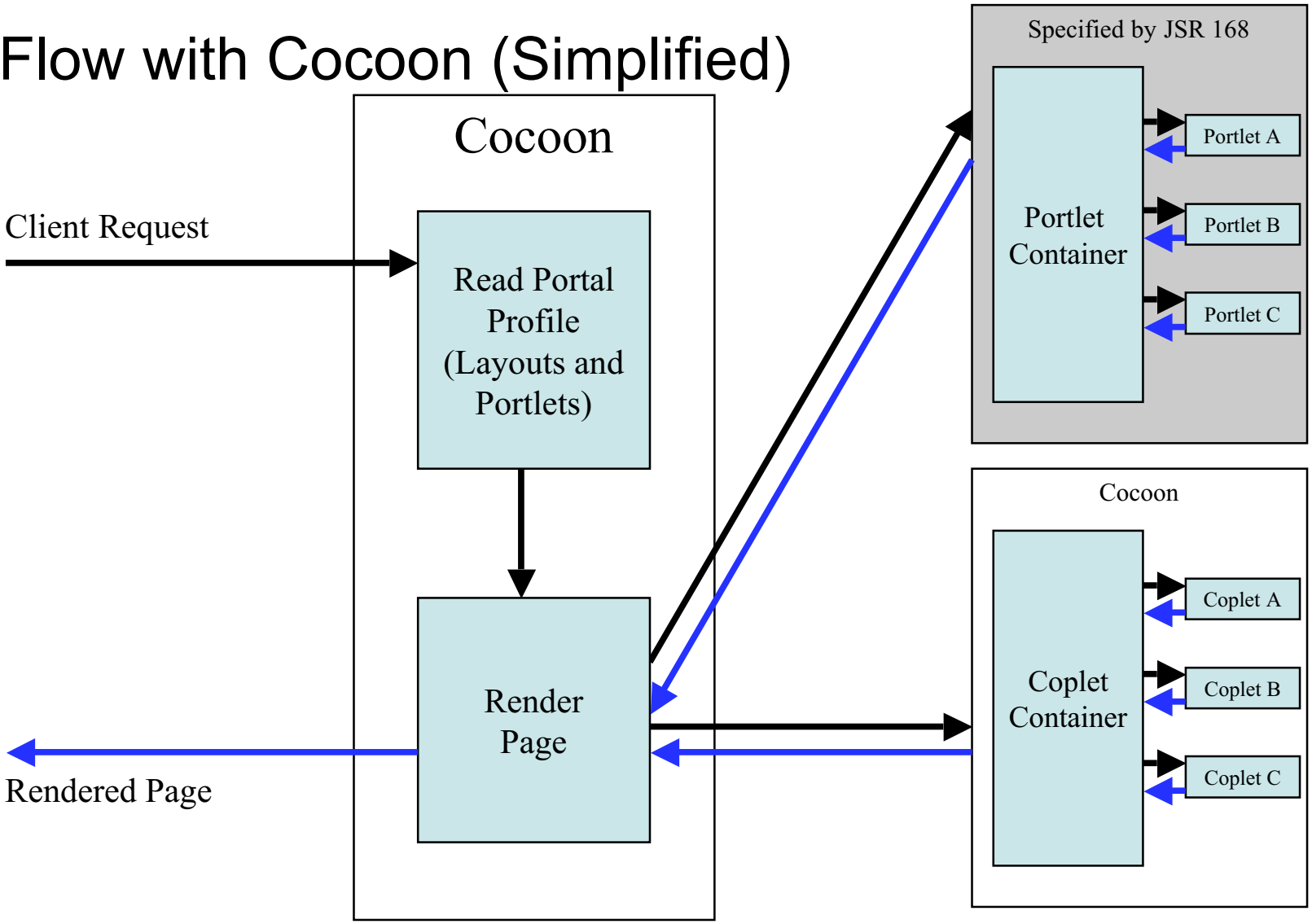
Request Response Cycle



Flow with JSR 168



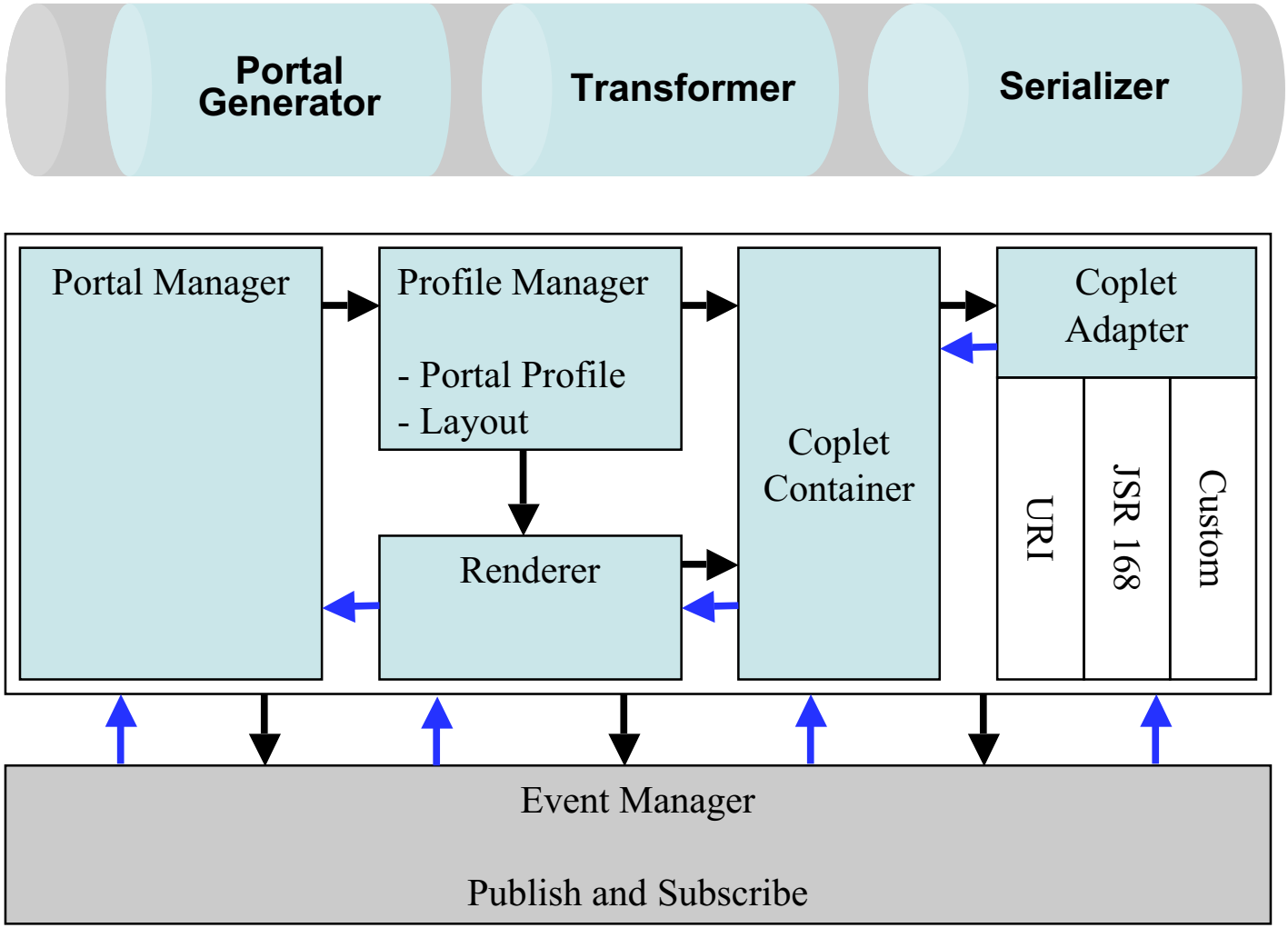
Flow with Cocoon (Simplified)



Plus: Event Handling

- Events for
  - Preference changes
  - Profile changes
  - Status changes
  - Inter-coplet communication

The Bigger Picture



The Bigger Picture

- The Cocoon Portal is
  - Assembled of several (Avalon) components
  - Change of behaviour through
    - Change of (XML) configuration
    - Exchanging/Extending components
  - Highly extensible
  - Pluggable

The Bigger Picture - Example

- The Profile Manager
  - Delivers the portal layout
  - Delivers the profile
  - Delivers preferences
  - Manages (persists) all of the above

The Bigger Picture - Example

- Static Profiles
  - Reads static XML files
- Profiles per User
  - Authentication Framework in Cocoon
    - Authentication of Users
    - Protection of Documents (Pipelines)
    - Flexible Integration of (existing) User Databases
  - One persistent profile for each user
    - XML documents
    - Flexible storaging anywhere
  - Profile Hierarchy
    - Role based
- Your own Profile Manager

Portal Features

- Everything is configurable
  - Available coplets
  - Features of a coplet
  - Layout information
  - User rights
- Flexible Presentation
  - Stylesheets for multi-channel
  - Coplets can be channel-dependent
- Parallel Processing

The Cocoon Portal

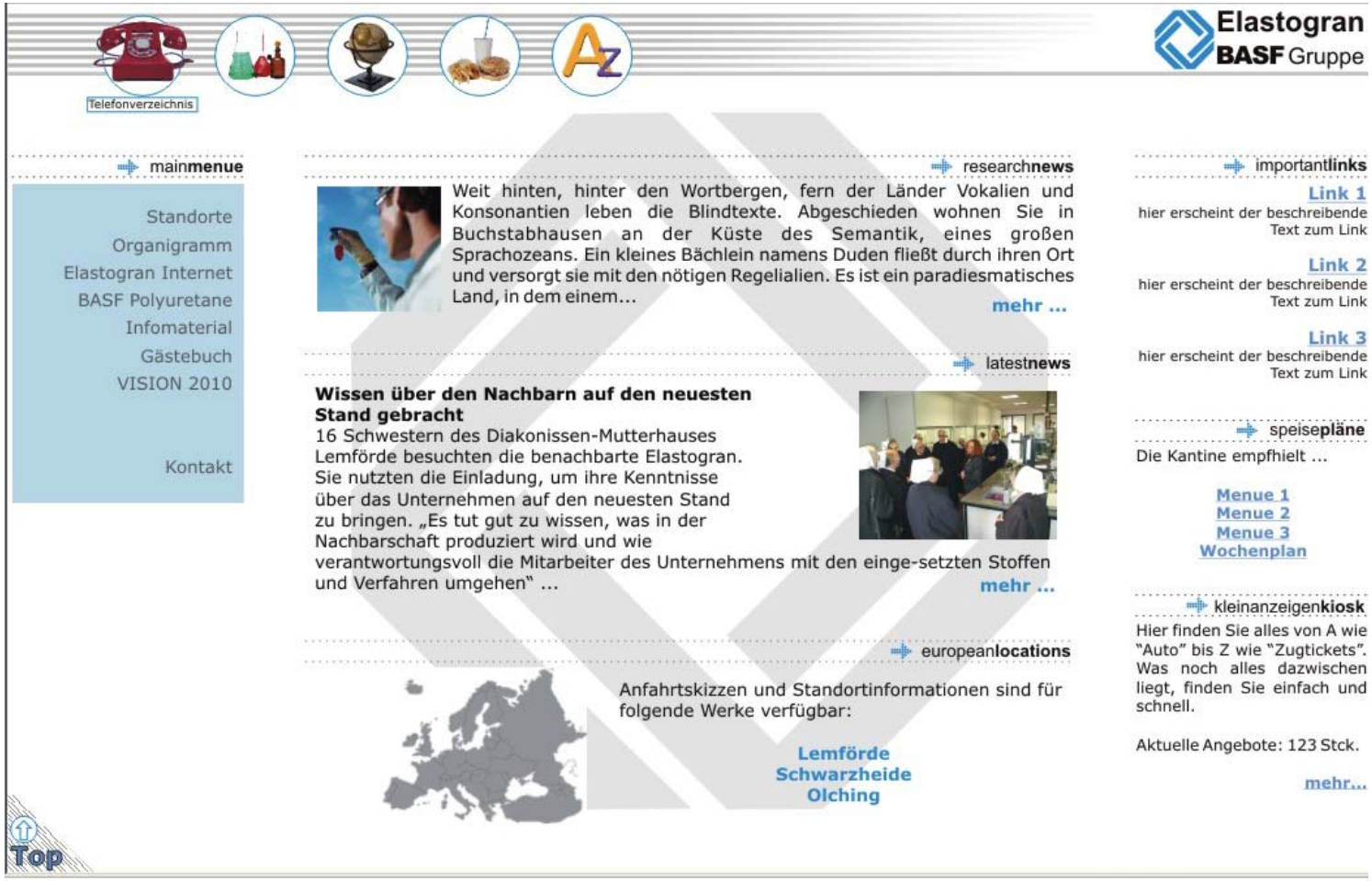
Real World Examples



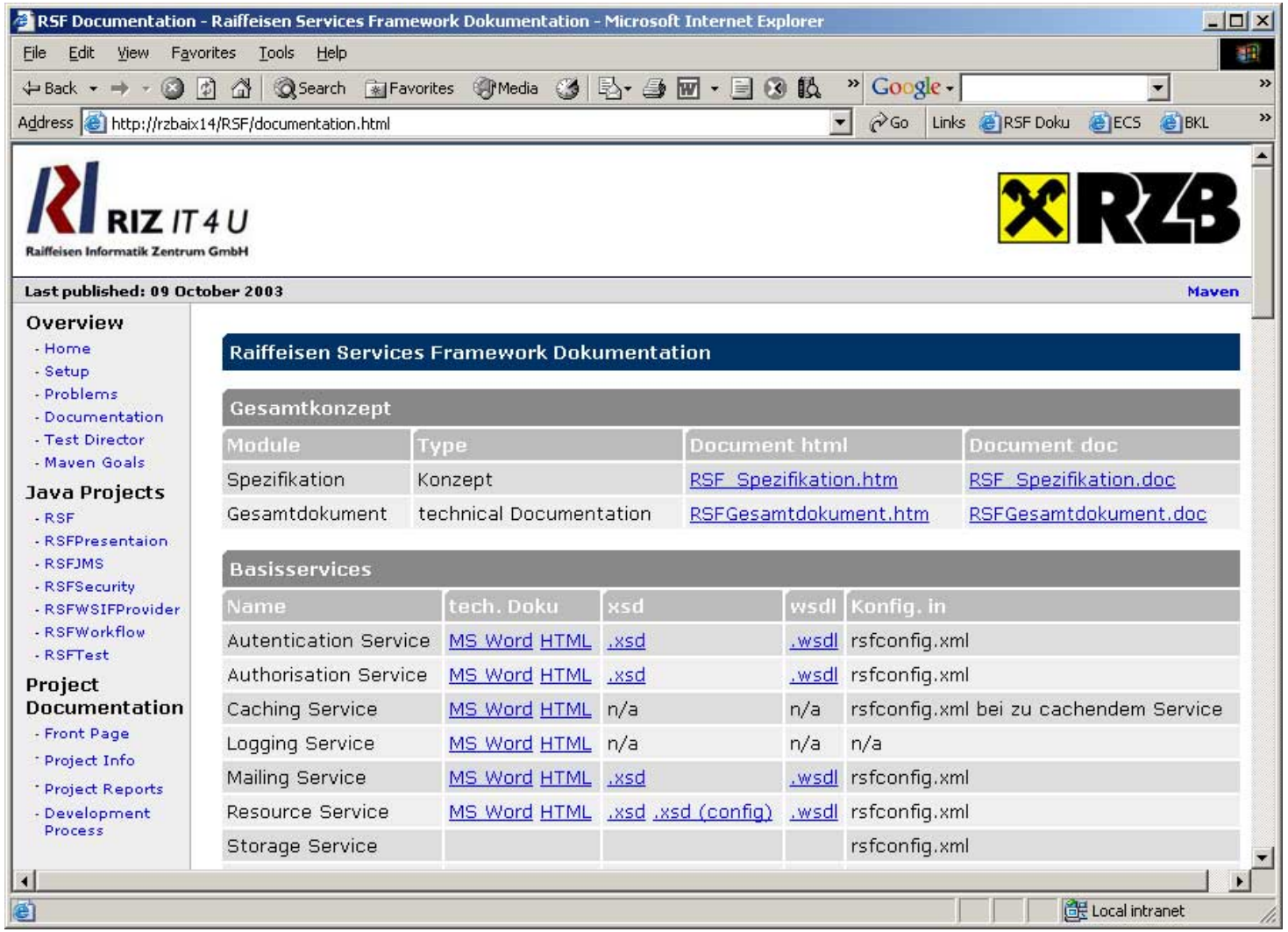
Portal Example at BASF



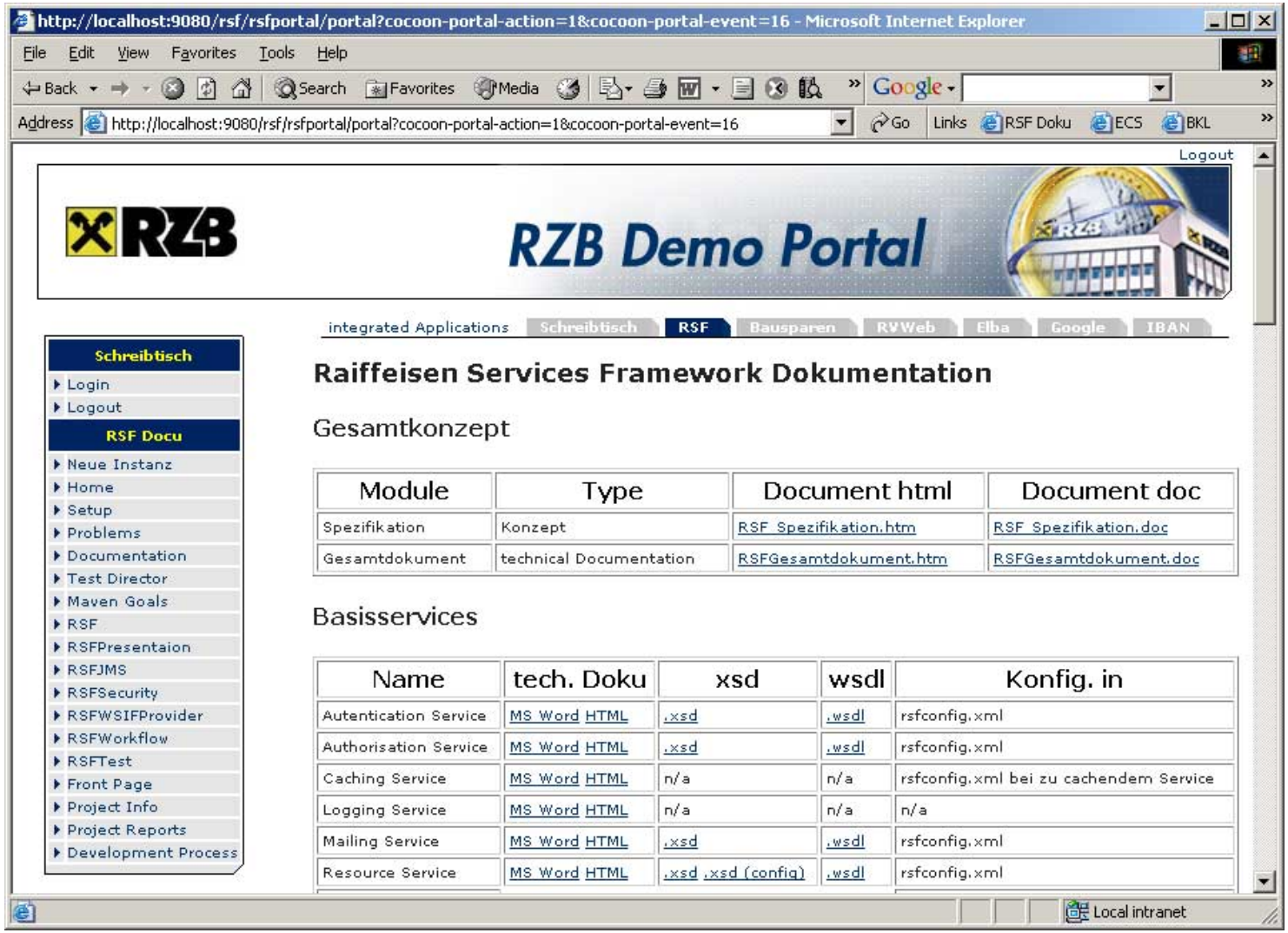
Portal Example at BASF



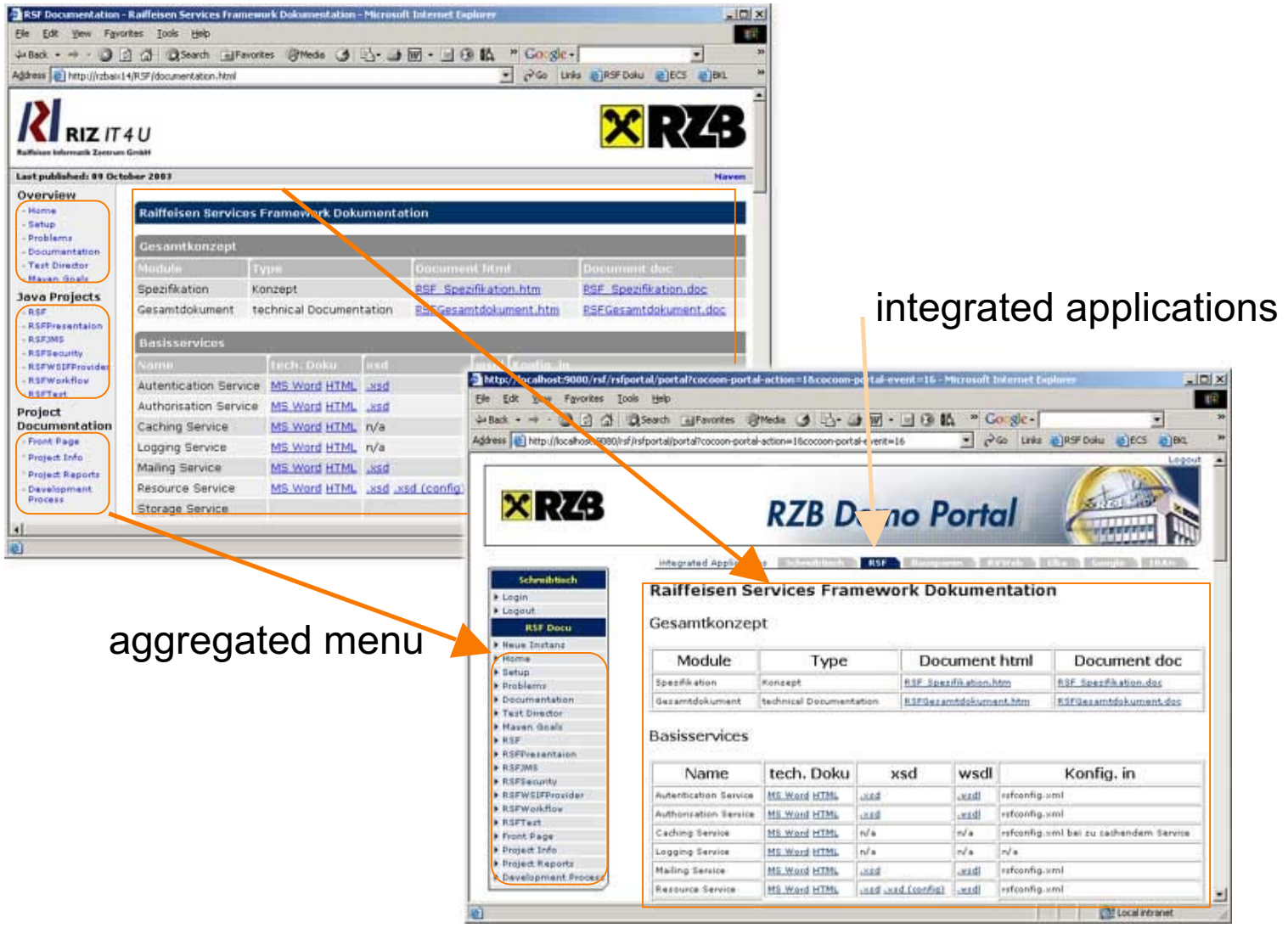
Integrating Application at RZB



Integrating Application at RZB



Integrating Application at RZB



The Cocoon Portal

Conclusion



The Cocoon Portal

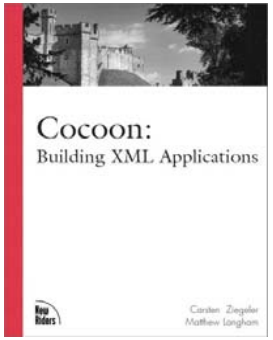
- Is tried and trusted
- All Cocoon features are usable
  - Flow
  - Form handling
- Usable without Java knowledge
- Extends the JSR 168
- Benefits from new concepts/features
  - In Cocoon (e.g. blocks)
  - **And** in the Portlet API

But Let's Be Honest…

- Initial version of (new) portal in Cocoon CVS
  - Including Demo
- Stable and usable
  - Documentation needed
- JSR 168 not implemented yet…
  - …but will be provided asap
- Will be expanded
- Join Us!
- And enjoy the great Cocoon Community!

Closing the Portal

- Cocoon and Portal
  - http://cocoon.apache.org
  - CVS checkout and build
- Tutorials
  - Check the Cocoon website for links
- The Book ☺
  - Cocoon: Building XML Applications, New Riders
  - ISBN:0735712352
- Cocoon training, presentations, projects (and more)
  - http://www.s-und-n.de
  - More information: ccos@s-und-n.de



Thanks for your attention!

Questions?

(And perhaps some answers ☺)



This presenation has ended.
Click anywhere to return.

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